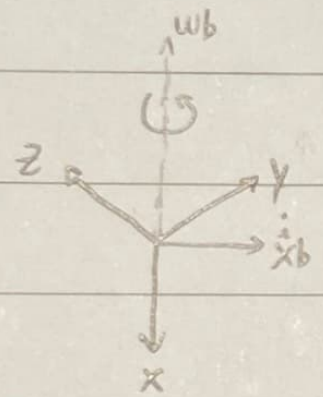
3.11. a) Sabe-se que $\dot{p}_i = \omega_i \times p_i$. Logo,

$$\dot{x}_b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{array}{ccc|cc} \hat{i} & \hat{j} & \hat{k} & \hat{i} & \hat{j} \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{array} = 0\hat{i} + \hat{j} + (-\hat{k}) = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\dot{y}_b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{array}{ccc|cc} \hat{i} & \hat{j} & \hat{k} & \hat{i} & \hat{j} \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{array} = -\hat{i} + 0\hat{j} + 0\hat{k} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$\dot{z}_b = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \hat{i} + 0\hat{j} + 0\hat{k} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$



b) Obtendo a matriz de rotação $Rot(\hat{z}, \theta)$, sendo

$$R_{eb} = Rot(\hat{z}, -90°) = \begin{bmatrix} \cos(-90°) & -\operatorname{sen}(-90°) & 0 \\ \operatorname{sen}(-90°) & \cos(-90°) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{eb} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\omega_e = R_{eb} \times \omega_b = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

c) Tendo $[\omega_s]R = \dot{R}$, obtém-se

$$\dot{R}_{eb} = [\omega_e] \cdot R_{eb} \rightarrow [\omega_e] = \begin{bmatrix} 0 & -\omega_{ez} & \omega_{ey} \\ \omega_{ez} & 0 & -\omega_{ex} \\ -\omega_{ey} & \omega_{ex} & 0 \end{bmatrix} \quad (I)$$

$$\dot{R}_{eb} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\dot{R}_{eb} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 0 \end{bmatrix}$$

d) Con base en (I):

$$[W_e] = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} ; \quad [W_b] = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$